

## Question

Using copper phthalocyanine and 2-methylimidazole as precursors, student Xiaoming directly synthesized a catalyst Cu@ZIF-8 via a hydrothermal reaction. Experimental studies have shown that this catalyst can reduce  $\text{CO}_2$  to products such as ethanol and acetone when a reduction potential is applied, but it also produces many reduction by-products, such as  $\text{H}_2$  and  $\text{CO}$ . To further investigate the performance of the catalyst, Xiaoming dropped 0.4mg of the catalyst onto a  $1\text{cm}^2$  carbon paper electrode and placed it in an electrolytic cell with a volume of 31mL. First, Xiaoming added 15mL of 0.5mol/L KOH solution to the electrolytic cell and purged it with Ar gas for 20min. After chronoamperometry electrolysis under sealed conditions, the mole fraction of  $\text{H}_2$  in the electrolytic cell was measured to be 1.584%, and the integral of the  $I-t$  curve of the chronoamperometry yielded a charge of 2.203C. Then, student Xiaoming purged  $\text{CO}_2$  gas for 60min under stirring conditions, and then switched to a closed system and performed chronoamperometry electrolysis. The mole fraction of  $\text{CO}$  in the electrolytic cell was measured to be 0.2110%, and the integral of the  $I-t$  curve of the chronoamperometry yielded a charge of 1.677C. Assuming the experimental conditions are room temperature and normal atmospheric pressure, what is the amount of substance of  $\text{CO}$  obtained during the test? Considering the gas tightness of the device, what should be the theoretical Faradaic efficiency of the  $\text{CO}$  by-product?

A.

$$1.380 \times 10^{-6} \text{ mol}$$

; 15.88 %

B.

$$1.380 \times 10^{-6} \text{ mol}$$

; 17.50 %

**C.**

$$1.380 \times 10^{-6} \text{ mol}$$

; 31.76 %

**D.**

$$2.761 \times 10^{-6} \text{ mol}$$

; 15.88 %

**E.**

$$2.761 \times 10^{-6} \text{ mol}$$

; 17.50 %

**F.**

$$2.761 \times 10^{-6} \text{ mol}$$

; 31.76 %

**G.** Insufficient device airtightness 90%, unable to calculate accurately

## Answer

Correct Answer: **B**

## Detailed Explanation

Faradaic efficiency calculation formula:

$$FE = \frac{n_{CO} \times z_{CO} \times F}{Q} \times 100\%$$

### CHECKPOINT

1 PTS

$$FE = \frac{n_{CO} \times z_{CO} \times F}{Q} \times 100\%$$

In this problem, the amount of substance of the gas needs to be calculated using the ideal gas law:

$$n_{CO} = \frac{p \times V}{R \times T}$$

## CHECKPOINT

1 PTS

$$n_{CO} = \frac{p \times V}{R \times T}$$

During the electrochemical reduction process, the reactions producing 1mol of H<sub>2</sub> and CO both involve the transfer of 2mol of electrons:



## CHECKPOINT

1 PTS

$$z_{\text{CO}} = z_{\text{H}_2} = 2$$

Analyzing the question, the volume of the gas container is  $31 - 15 = 16\text{mL}$ , the mole fraction of CO is 0.211%, the temperature is 298.15K, the pressure is 101.325kPa, R is 8.314J/mol · K, F is 96485C/mol, and the measured Q is 1.677C.

Therefore:

$$n_{CO} = \frac{101.325 \times 16 \times 0.2110 / 100000}{8.314 \times 298.15} = 1.380 \times 10^{-6}$$

mol

#### CHECKPOINT

1 PTS

$$n_{CO} = 1.380 \times 10^{-6} \text{ mol}$$

However, the question emphasizes the importance of the airtightness of the device, and the previous experiment with Ar atmosphere in KOH was to test this. Since only  $H_2$  is produced under these conditions, the theoretical FE should be 100% if there is no leakage, which is a common method for calibrating airtightness in electrochemistry.

#### CHECKPOINT

3 PTS

To be precise, under Ar protection and KOH electrolyte conditions, the electroreduction reaction can only produce hydrogen, so the Faradaic efficiency of hydrogen can be used to test the airtightness of the device

$$n_{H_2} = \frac{101.325 \times 16 \times 1.584 / 100000}{8.314 \times 298.15} = 1.036 \times 10^{-5} \text{ mol}$$

$$FE_{H_2} = \frac{1.036 \times 10^{-5} \times 2 \times 96485}{2.203} = 90.74\%$$

The airtightness of the device is greater than 90%, and the overall airtightness is good.

#### CHECKPOINT

2 PTS

$$FE_{H_2} = 90.74\% > 90\%$$

, good airtightness

Therefore, the correction factor is obtained:

$$\eta = 1/0.9074 = 1.102$$

#### CHECKPOINT

2 PTS

Correction factor  $\eta = 1.102$

The corrected ideal  $CO$  amount:

$$n_{CO} = 1.380 \times 10^{-6} \times 1.102 = 1.521 \times 10^{-6}$$

mol

CHECKPOINT

2 PTS

Corrected

$$n_{CO} = 1.521 \times 10^{-6} \text{ mol}$$

The final true Faradaic efficiency:

$$FE = \frac{1.521 \times 10^{-6} \times 2 \times 96485}{1.677} = 17.50\%$$

CHECKPOINT

2 PTS

Corrected

$$FE_{CO} = 17.50\%$$